

ID1063: Introduction to Programming
Aug-Nov 2023
IIT-Hyderabad
Mid-Semester Exam

October 14, 2023

Answer the questions in the spaces provided. If you run out of room, ask for an answer booklet. Read the questions carefully before answering. All the best!

Name: _____

Roll No: _____

Section 1. Answer True/False ($6 \times 1 = 6$ marks)

- _____ The “&&” operator in C performs a logical OR operation.
- _____ Compiler generates object code that is executed directly by the machine.
- _____ A variable declared with the **static** keyword in a function retains its value across all functions of a program.
- _____ Both **#define** and **#include** are preprocessor directives.
- _____ The **-f** command in **gcc** specifies the filename of the compilation output.
- _____ A **do-while** loop is guaranteed to execute its block of code at least once.

Section 2. Multiple Choice Questions ($8 \times 1 = 8$ marks)

1. Consider the following code snippet, and state the expected output:

```
int x = 5;
printf("%d",x++);
x += 3;
printf("%d",x);
```

- (a) 6,9
- (b) 5,9
- (c) 5,8
- (d) 6,8
2. Which of the following is NOT a valid data type in C?
- (a) string
- (b) int
- (c) float
- (d) double
3. Which among the following operators have the highest precedence in C?
- (a) *
- (b) %
- (c) =
- (d) ++

4. You have defined an integer array `A` of size 1. You are assigning a value 1 to `A[1]`. What type of error is this?
- (a) No error
 - (b) Compiler error
 - (c) Cannot define an array of size 1
 - (d) None of the above
5. Which one of the following statements is **true**?
- (a) The stack frame corresponding to a function under execution does not contain the global variables used by the function.
 - (b) Function prototype needs to list the names of the function argument variables.
 - (c) Function call stack contains frames corresponding to all functions defined in the program.
 - (d) Variable reusability is an important advantage of having function calls.
6. Which of the following is not a valid way to declare an array in C?
- (a) `int numbers[5];`
 - (b) `int[5] numbers;`
 - (c) `int[] numbers = 1, 2, 3, 4, 5;`
 - (d) `int numbers[] = 1, 2, 3, 4, 5;`
7. Which one of the following statements is **false**?
- (a) `double` can store a value of much greater magnitude with greater precision than `float`
 - (b) `scanf` returns the number of items that were read
 - (c) The `default` case is optional in a `switch-case` statement.
 - (d) A `for` loop statement must always contain an *initialization*, a *condition*, and an *increment/decrement/update to the loop variable*.
8. Which of the following statements is **false**?
- (a) Arithmetic expressions with mixed-data-types invoke implicit type casting
 - (b) Casting a `short int` to `float` is not a safe operation
 - (c) Type casting is used when the value to be stored is in a form not supported by that data type
 - (d) More than one explicit casting can be used in a single C statement

Section 3. Predict the outputs ($8 \times 2 = 16$ marks)

For each code snippet, assume that the appropriate header files are already included.

Add a one-line justification for your answer.

```
1. void main()
{
    int y, x, c; float m;
    y=5; x=4; c=3;
    printf("m=%f", (y-c)/x);
}
```

```
2. void main()
{
    int i=1;
    i=i++ + ++i;
    printf("%d", i);
}
```

```
3. void main()
{
    int v1=10, v2=-20, negnum=0;

    if(v1*v2 < -1) {
        int negnum=1;
    }

    if (negnum == 1) {
        printf("One number is negative");
    }
    else {
        printf("Both are positive or both are negative");
    }
}
```

```
4. #include<stdio.h>
enum {ONE=1, TWO=2}

void main()
{
    printf("%d", ONE+TWO);
}
```

```
5. void main()
{
    int i=rand();
    printf("%d", (i*(i+1))%2);
}
```

```
6. void main()
{
    int i, a[3]={0};
    for (i=0; i<3; i++) {
        ++a[i++];
    }
    printf("%d,%d,%d",a[0], a[1], a[2]);
}
```

```
7. void main()
{
    int i=1, j=2;
    i=i%2, j=j%2;
    if(i=0) {
        printf("%d", j);
    }
    else {
        printf("%d", i);
    }
}
```

```
8. void main()
{
    int x=y=1;
    printf("%d",x-y);
}
```

Answer Key for Exam A

Section 1. Answer True/False ($6 \times 1 = 6$ marks)

- False The “&&” operator in C performs a logical OR operation.
- False Compiler generates object code that is executed directly by the machine.
- False A variable declared with the **static** keyword in a function retains its value across all functions of a program.
- True Both **#define** and **#include** are preprocessor directives.
- False The **-f** command in **gcc** specifies the filename of the compilation output.
- True A **do-while** loop is guaranteed to execute its block of code at least once.

Section 2. Multiple Choice Questions ($8 \times 1 = 8$ marks)

1. Consider the following code snippet, and state the expected output:

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- (c) float
- (d) double
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- (d) ++
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- (d) None of the above
5. Which one of the following statements is **true**?
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- (b) Function prototype needs to list the names of the function argument variables.
- (c) Function call stack contains frames corresponding to all functions defined in the program.
- (d) Variable reusability is an important advantage of having function calls.

6. Which of the following is not a valid way to declare an array in C?

- (a) `int numbers[5];`
- ☒ (b) `int[5] numbers;`
- (c) `int[] numbers = 1, 2, 3, 4, 5;`
- (d) `int numbers[] = 1, 2, 3, 4, 5;`

7. Which one of the following statements is **false**?

- (a) `double` can store a value of much greater magnitude with greater precision than `float`
- (b) `scanf` returns the number of items that were read
- (c) The `default` case is optional in a `switch-case` statement.
- ☒ (d) A `for` loop statement must always contain an *initialization*, a *condition*, and an *increment/decrement/update to the loop variable* .

8. Which of the following statements is **false**?

- (a) Arithmetic expressions with mixed-data-types invoke implicit type casting
- ☒ (b) Casting a `short int` to `float` is not a safe operation
- (c) Type casting is used when the value to be stored is in a form not supported by that data type
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Section 3. Predict the outputs ($8 \times 2 = 16$ marks)

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Add a one-line justification for your answer.

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1. void main()
{
    int y, x, c; float m;
    y=5; x=4; c=3;
    printf("m=%f", (y-c)/x);
}
```

Answer: `m=0.000000`

```
2. void main()
{
    int i=1;
    i=i++ + ++i;
    printf("%d", i);
}
```

Answer: `4`

```

3. void main()
{
    int v1=10, v2=-20, negnum=0;

    if(v1*v2 < -1) {
        int negnum=1;
    }

    if (negnum == 1) {
        printf("One number is negative");
    }
    else {
        printf("Both are positive or both are negative");
    }
}

```

Answer: Both are positive or both are negative

```

4. #include<stdio.h>
enum {ONE=1, TWO=2}

void main()
{
    printf("%d", ONE+TWO);
}

```

Answer: Compile Error. Semicolon missing after enum declaration.

```

5. void main()
{
    int i=rand();
    printf("%d", (i*(i+1))%2);
}

```

Answer: 0

```

6. void main()
{

```



```

    int i, a[3]={0};
    for (i=0; i<3; i++) {
        ++a[i++];
    }
    printf("%d,%d,%d",a[0], a[1], a[2]);
}

```

Answer: 1,0,1

7. void main()
{
 int i=1, j=2;
 i=i%2, j=j%2;
 if(i=0) {
 printf("%d", j);
 }
 else {
 printf("%d", i);
 }
}

Answer: 0

8. void main()
{
 int x=y=1;
 printf("%d",x-y);
}

Answer: Compile Error.